

Algorithmic Pseudocode Sample 1

Source-backed Image2Struct algorithm sample

1 Algorithm

This sample contains algorithmic pseudocode extracted from a source-backed LaTeX benchmark dataset. It is wrapped in a minimal article document for pdfLaTeX validation.

Algorithm 1 Source-backed algorithmic procedure

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for  $t \in \{-1, \dots, -T^{traceback}\}$  do                                 $\triangleright$  Initialization of  $\delta_t^{turned\_on}$  and  $\delta_t^{turned\_off}$ 
     $\delta_t^{turned\_on} \leftarrow 0$ 
     $\delta_t^{turned\_off} \leftarrow 0$ 
    if  $S_{u,t}^{STOP} - S_{t-1}^{STOP} = 1$  then                                 $\triangleright$  Replace by  $S_{u,t}^{OFF} - S_{t-1}^{OFF} = 1$  if STOP is not defined.
         $\delta_t^{turned\_off} \leftarrow 1$ 
    else if  $S_{u,t}^{START} - S_{t-1}^{START} = 1$  then                         $\triangleright$  Replace by  $S_{u,t}^{OFF} - S_{t-1}^{OFF} = -1$  if START is not
defined.
         $\delta_t^{turned\_on} \leftarrow 1$ 
    end if
end for
for  $t \in \{-1, \dots, -T^{traceback}\}$  do                                 $\triangleright$  Initialization of  $\delta_t^{stable}$ ,  $\delta_t^{entered\_up}$  and  $\delta_t^{entered\_down}$ 
     $\delta_t^{stable} \leftarrow 0$ 
     $\delta_t^{entered\_up} \leftarrow 0$ 
     $\delta_t^{entered\_down} \leftarrow 0$ 
    if  $S_{u,t}^{STOP} - S_{t-1}^{STOP} = 1$  then
         $\delta_t^{stable} \leftarrow 1$ 
    else if  $S_{u,t}^{ON\_UP} - S_{t-1}^{ON\_UP} = 1$  then
         $\delta_t^{entered\_up} \leftarrow 1$ 
    else if  $S_{u,t}^{ON\_UP} - S_{t-1}^{ON\_UP} = 1$  then
         $\delta_t^{entered\_down} \leftarrow 1$ 
    end if
end for
for  $t \in \{-1, \dots, -T^{traceback}\}$  do                                 $\triangleright$  Initialization of  $\delta_t^{stable}$ ,  $\delta_t^{entered\_up}$  and  $\delta_t^{entered\_down}$ 
     $\delta_t^{stable} \leftarrow 0$ 
     $\delta_t^{entered\_up} \leftarrow 0$ 
     $\delta_t^{entered\_down} \leftarrow 0$ 
    if  $S_{u,t}^{ON\_FLAT} - S_{t-1}^{ON\_FLAT} = 1$  then
         $\delta_t^{stable} \leftarrow 1$ 
    else if  $S_{u,t}^{ON\_UP} - S_{t-1}^{ON\_UP} = 1$  then
         $\delta_t^{entered\_up} \leftarrow 1$ 
    else if  $S_{u,t}^{ON\_DOWN} - S_{t-1}^{ON\_DOWN} = 1$  then
         $\delta_t^{entered\_down} \leftarrow 1$ 
    end if
end for
for  $t \in \{-1, \dots, -T^{traceback}\}$  do                                 $\triangleright$  Initialization of  $\delta_t^{flat,down,stop}$  or  $\delta_t^{down\_to\_stop}$ 
     $\delta_t^{flat,down,stop} \leftarrow \left\lfloor \frac{S_{u,t}^{STOP} + S_{t-1}^{ON\_DOWN} + S_{t-2}^{ON\_FLAT}}{3} \right\rfloor$ 
     $\delta_t^{down\_to\_stop} \leftarrow 0$ 
    if  $S_{u,t}^{STOP} - S_{t-1}^{ON\_DOWN} = 0$  then
         $\delta_t^{down\_to\_stop} \leftarrow 1$ 
    end if
end for
 $U_{-1} = S_{-1}^{ON\_UP} \times S_{-2}^{ON\_UP} \times (P_{u,t-1} - P_{u,t-2})$                                  $\triangleright$  Initial condition on  $U_t$ 
 $D_{-1} = S_{-1}^{ON\_DOWN} \times S_{-2}^{ON\_DOWN} \times (P_{u,t-1} - P_{u,t-2})$                                  $\triangleright$  Initial condition on  $D_t$ 

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